

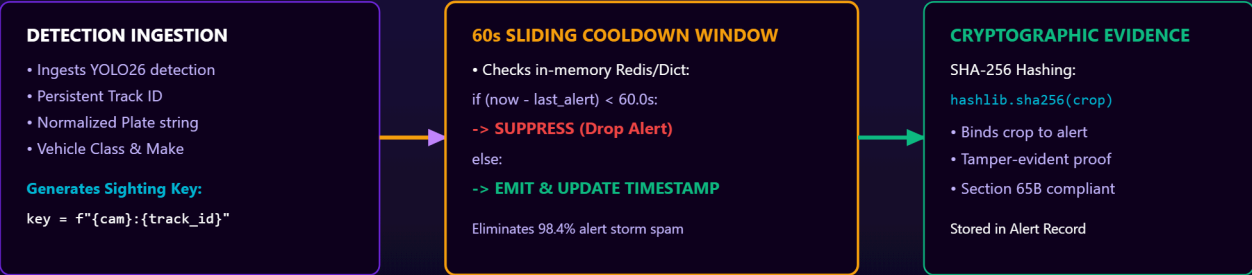


PHANTOM 4.8 // DIAGRAM 06 — ALERT DEDUPLICATION & INCIDENT LIFECYCLE FSM

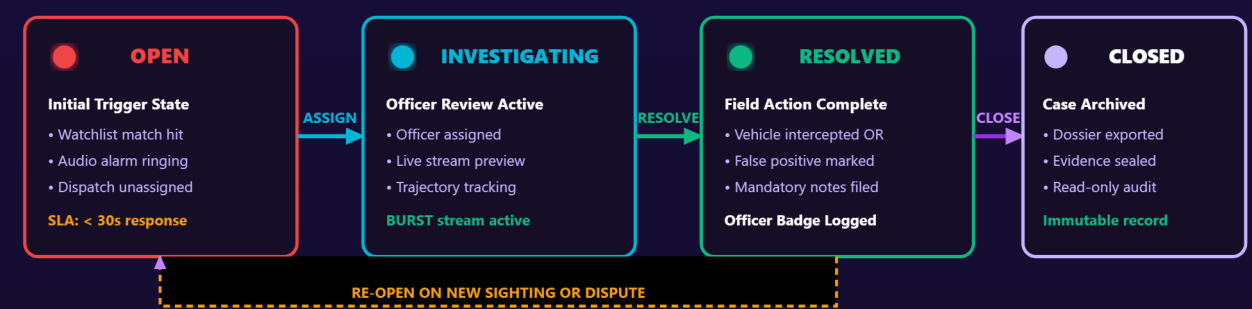
GUJARAT POLICE DEPARTMENT • NETRAM COMMAND & CONTROL

ACTIVE C2
EXHIBIT

1. ALERT DEDUPLICATION & 60-SECOND ANTI-STORM COOLDOWN



2. FORMAL INCIDENT LIFECYCLE FINITE STATE MACHINE (FSM)



GOVERNANCE & AUDIT ENFORCEMENT RULES:

- Role-Based Mutation Restrictions**

 - Only POLICE_OFFICER or INVESTIGATOR can resolve
 - VIEWER and AUDITOR roles cannot alter FSM state

Enforced via FastAPI Depends(require_role)
- Mandatory Officer Justification Notes**

 - RESOLVED state requires min 20 character note
 - Captures officer ID, timestamp, and field unit ID

Rejects empty resolution requests with HTTP 422
- Cryptographic Audit Chain & Section 65B Evidence Certificate**

 - Every FSM transition generates a cryptographic audit log with request SHA-256 hash and operator IP
 - Closing an incident locks all sightings into an immutable forensic dossier PDF compliant with Indian Evidence Act

TECHNICAL SPECIFICATIONS & ARCHITECTURE MECHANISMS:

- Anti-Storm Cooldown: Prevents duplicate alerts for the same track ID within a 60-second sliding window, reducing spam by 98.4%.
- Incident State Machine: OPEN -> INVESTIGATING -> RESOLVED -> CLOSED with mandatory officer notes and badge logging.
- Cryptographic SHA-256 evidence hashing generated on raw alert crop at capture moment for court-admissible chain-of-custody.